

6.6 Designing a Data-Collection Approach

ڈیٹا جمع کرنے سے پہلے منصوبہ بندی ضروری ہے: تحقیقی سوال واضح کریں، ڈیٹا کی قسم طے کریں، مناسب طریقہ چنیں، نمونہ...

CS XI

Chapter 6: Digital Literacy

Lecture 07 of 14

6.6 Designing a Data-Collection Approach

LEARN IT

Before collecting data, a researcher must have a **plan**. A plan makes sure that the information gathered is **accurate**, **relevant** (related to the question) and **useful**. Follow these eight steps in order:

1. **Define the research question clearly.** What exactly do you want to find out?
2. **Decide the type of data needed:** quantitative, qualitative, or both.
3. **Select suitable methods** (survey, interview, observation).
4. **Choose the sample or target group** – the people who will give you data.
5. **Prepare the tools** such as forms, questions and observation sheets.
6. **Collect the data** carefully and accurately.
7. **Store the data digitally** in a spreadsheet and **keep a backup**.
8. **Follow ethical practice and respect privacy. Consent is mandatory** (you must get permission).

Question → Data type → Method → Sample → Tools → Collect → Store + backup → Ethics

WORKED EXAMPLE

Applying the eight steps. Topic: water wastage in your college.

1. **Question:** "How much water is wasted at the college taps during break time, and why?"
2. **Data type:** Both – quantitative (number of taps left running, minutes) and qualitative (reasons students give).
3. **Methods:** Observation (count running taps) + a short survey (ask about habits).
4. **Sample:** All 12 taps in the college; 30 students from different classes for the survey.
5. **Tools:** An observation sheet with columns *Tap no.* / *Time* / *Left running?* (Y/N) / *Minutes*; a Google Form with 5 questions.
6. **Collect:** Observe during break for five days; send the form link.
7. **Store:** Enter the observation counts into Google Sheets; download a copy to a USB drive as backup.
8. **Ethics:** Ask the principal's permission; tell students the survey is anonymous and voluntary; do not record names.

REMEMBER IT

Remember the steps with: "**Q**uiet **T**eachers **M**ake **S**tudents **T**hink, **C**heck, **S**ave **E**thically" → Question, Type, Method, Sample, Tools, Collect, Store, Ethics.

COMMON MISTAKE

Mistake: Starting to collect data before writing a clear research question. You then collect lots of data that does not answer anything. **Fix:** Always write the question first. Another mistake: keeping data in only one place – if the file is lost, the project is lost. Always keep a backup.

اردو میں سمجھیں

ڈیٹا جمع کرنے سے پہلے منصوبہ بندی ضروری ہے: تحقیقی سوال واضح کریں، ڈیٹا کی قسم طے کریں، مناسب طریقہ چنیں، نمونہ (Sample) منتخب کریں، فارم اور سوالات تیار کریں، احتیاط سے ڈیٹا جمع کریں، اسے اسپریڈشیٹ میں محفوظ کر کے بیک اپ رکھیں، اور اخلاقیات کا خیال رکھیں۔ اجازت (Consent) لینا لازمی ہے۔

CHECK YOURSELF

1. What is the first step in designing a data-collection approach?
2. Which step says "consent is mandatory"?
3. Where should collected data be stored?

Answers: 1. Define the research question clearly. 2. Step 8 – follow ethical practice and respect privacy. 3. Digitally, in a spreadsheet, with a backup copy.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Data-collection strategies include:

- a. Interview, survey, prototype, observation, simulation
- b. Only NAND
- c. Only OSI ping
- d. Only K-maps

MCQ 2. You collect data when:

- a. Existing sources are not enough
- b. You already have the full census always
- c. The question is undefined
- d. You only need a meme

MCQ 3. A strategy must match:

- a. The question and the people
- b. The teacher's favourite chart only
- c. Agile always
- d. TCP window

MCQ 4. Bias in collection:

- a. Systematically distorts answers
- b. Improves SD always
- c. Is required in exams
- d. Is a primary key

MCQ 5. Ethics: people should:

- a. Know and consent
- b. Be filmed secretly always
- c. Hand over passwords
- d. Be named in viral posts

MCQ 6. This is ★ because papers ask:

- a. Name + when to use + one limit of a method
- b. Only Python turtles
- c. Only ER
- d. Only Big O

Extra short questions

1. List five strategies from the book.
2. One advantage of planning methods first.
3. One ethical rule.
4. What is a biased question?

Extra long question

1. Choose methods for: (a) why students skip practicals (b) how a new ID-card prototype feels (c) traffic at the gate. Justify. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Interview, survey, prototype, observation, simulation (and others in 6.4).
2. You do not collect the wrong type of data for your chart later.
3. Informed consent; anonymise students.
4. "Don't you agree our canteen is terrible?" — pushes the answer.

Long

1. (a) interview+survey (b) prototype test (c) observation/count or simulation. Limits of each. Ethics sentence.